

Section 1			
Question	Question title	Marks	Question type
i	Info - DRAFT		Information or resources
1	Q	1	Multiple Choice
2	Q	1	Multiple Choice
3	Q	1	Multiple Choice
4	Q	1	Multiple Choice
5	Q	1	Multiple Choice
6	Q	1	Multiple Choice
7	Q	1	Multiple Choice
8	Q	1	Multiple Choice
9	Q	1	Multiple Choice
10	Q	1	Multiple Choice
11	Q	1	Multiple Choice
12	Q	1	Multiple Choice
13	Q	1	Multiple Choice
14	Q	1	Multiple Choice
15	Q	1	Multiple Choice
16	Q	1	Multiple Choice
17	Q	1	Multiple Choice
18	Q	1	Multiple Choice
19	Q	1	Multiple Choice
20	Q	1	Multiple Choice

i Info - DRAFT

Exam: DAT540-1 22H Introduksjon til datavitenskap

Available time: ___ hrs for submitting hh:mm-hh:mm hrs

All exams submitted after the expired upload time will not be registered.

Questions - All Questions will be graded after the exam. Therefore, you will NOT receive any points immediately after completion.

- **Multiple Choice (40 questions):** You get 1 point for each correct answer to the multiple choice questions. You will get 0 points for incorrect answers or unattempted questions.
- **File Uploads / Essay Type Questions: NONE**

Supporting material: All technical support material is permitted. However, you are not allowed to get help from other people when working on your exam assignment. We are also reminding you that you when registering for the semester, you signed that you have read and understood the rules for cheating and plagiarism in the Exam Rules and Regulations at the University of Stavanger. Plagiarism control will be carried out.

Important contact information during the exam: If something should happen during the exam and you need help, you can call one of the phone numbers below if you need technical support, administrative support, or have questions about the course responsible.

Course responsible:

- Antorweep Chakravorty, antorweep.chakravorty@uis.no, Phone: +4740625773
- Neel Kanwal, neel.kanwal@uis.no, Phone: +47 ...

Administrative support: 51 83 31 26

NB! If something goes wrong in Inspira and you cannot hand in your assignment, you need to contact administrative support ASAP.

Important Information about Exams in Inspira

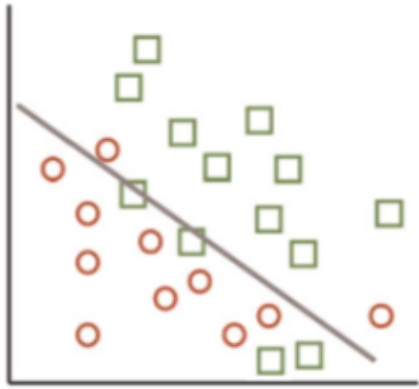
If you have been given extra time on the exam, it has been added. You can see the count down at the top of the page. Here you will also find your candidate number.

Handing in

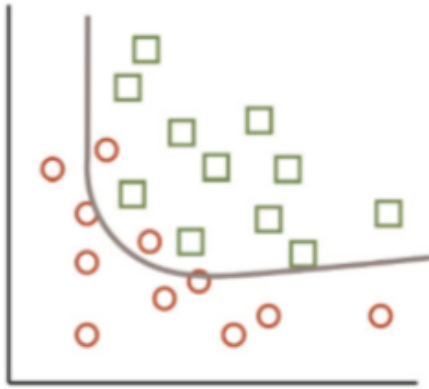
The exam will automatically close when the time is up. We recommend that you get yourself informed of how to best carry these steps out in due time and before you start working on the assignments.

1 Q

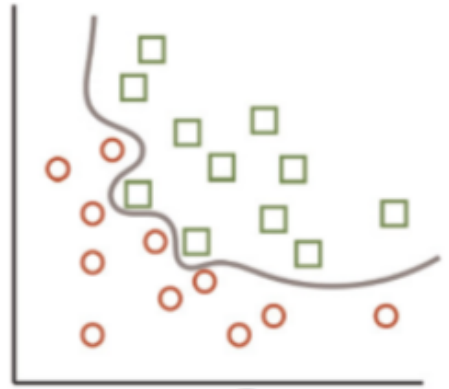
Which of the following model learning curves are undesirable (Not a good fit for generalization error)?



1



2



3

Select one alternative:

☐ 2 & 3

☐ 1

☐ 1 & 3

☐ 3



Maximum marks: 1

What is the output of the following code

```
a = np.array([[3,7,9],  
              [13, 2, 4]])  
b = np.array([[6,5,4],  
              [4,5,6]])  
[i>j for i,j in zip(a,b)][0]
```

Select one alternative:

- ☒ array([False, True, True])
- ☐ array([True, False, False])
- ☐ [array([False, True, True]), array([True, False, False])]
- ☐ [array([False, False, True]), array([True, False, False])]



Maximum marks: 1

3

Q

A machine learning model with a high bias might have
Select one alternative:

- ☐ Too many model parameters
- ☐ None of these
- ☐ High sensitivity to new samples
- ☐ Low variance



Maximum marks: 1

Assuming two data frames, A and B, which of the following statement can correctly join them horizontally?

A

	A	B
0	5	11
1	16	22

B

	C	D
0	15	1
1	18	33

Select one alternative:

- ☐ `pd.concat([a, b], axis=1)`
- ☐ `a.combine_first(b)`
- ☐ `a.join(b)`
- ☐ All of these



Maximum marks: 1

5

Q

What would the following code print when num=10?

```
def test(num):  
    try:  
        if num%2!=0:  
            return num  
        else:  
            return num+1  
    except:  
        return num-2  
k = test(10) + 1  
print(k)
```

Select one alternative:

- ☐ 10
- ☐ 11
- ☐ 12
- ☐ None of these



Maximum marks: 1

6

Q

Given a list, `a = [1,2,3,4,5,6,7,8,9]`, what would be the output of `a[::-3][::-1]`

Select one alternative:

☐ [7, 5, 3]

☐ [3, 5, 7]

☐ [9, 6, 3]

☐ [3, 6, 9]



Maximum marks: 1

What does this function return?

```
def whatItReturns(argument):  
    if argument == 1:  
        return 0  
  
print(whatItReturns(0))
```

Select one alternative:

- ☐ None of the above
- ☐ Just returns 0
- ☐ Returns NoneType Object
- ☐ Returns nothing and the program exits



Maximum marks: 1

8

Q

Given the dataframe,

```
df = pd.DataFrame({'first': ['a', 'b'],  
                  'second': ['c', 'd']})
```

what would be the datatype of the output after the application of pipe function and np.sum inside of it.

```
type(df.pipe(np.sum))
```

Select one alternative:

- ☐ the pandas dataframe itself
- ☐ the pandas series
- ☐ the code would generate an error
- ☐ the numpy ndarray



Maximum marks: 1

9 Q

Given the numpy array, what would be the outputs as listed in the code

```
nparray = np.arange(8).reshape(2,2,2)
nparray.size
len(nparray)
nparray.shape
```

Select one alternative:

- ☐ 8, 8 and (2, 2, 2)
- ☐ 2, 8 and (2, 2, 2)
- ☐ 8, 2 and (2, 2, 2)
- ☐ 2, 2 and (2, 2, 2)



Maximum marks: 1

Given the following code, what would be the output,

```
first = np.array([[1, 2, 3],  
                  [2, 5, 8]])  
second = first.copy()  
np.concatenate((first,second), axis=0)
```

OUTPUTS ->

a).

```
array([[1, 2, 3],  
       [2, 5, 8],  
       [1, 2, 3],  
       [2, 5, 8]])
```

b).

```
array([[1, 2, 3, 1, 2, 3],  
       [2, 5, 8, 2, 5, 8]])
```

c).

```
array([[1, 2, 3],  
       [1, 2, 3],  
       [2, 5, 8],  
       [2, 5, 8]])
```

d). None.

Select one alternative:

☐ d

☐ a

☐ c

☐ b



Maximum marks: 1

You have imported datetime library from datetime module and generated a datetime object,
`date_time = 2022-12-23 14:15:00`

Now if you implement the following piece of code, what will be the value of `date_time`?

```
date_time.replace(minute=30)  
date_time
```

Select one alternative:

- ☐ 2022-12-23 14:30:00
- ☐ 2022-12-23 14:15:00
- ☐ None of the these
- ☐ 2022-12-23 14:45:00



Maximum marks: 1

We declared a tuple, `tuple = ( [97, 56, 70], 'data science', (4, 20))`. What will happen if we want to append '12' at `tuple[0]` and `tuple[2]`?

Select one alternative:

- ☐ A. `([97, 56, 70, '12'], 'data science', (4, 20, '12'))`
- ☐ B. `([97, 56, 70, '12'], 'data science', (4, 20, 12))`
- ☐ It will throw an error
- ☐ Option A and B



Maximum marks: 1

We have, `li = {1 : 'one', 2 : 'two', 3 : 'three', 4: 'four'}`. Which of the following implementations will generate the following output?

1 2 3 4

OUTPUTS ->

a).

```
for i, value in enumerate(li):  
    print(value)
```

b).

```
for i, value in li.items():  
    print(i)
```

c).

```
for i in li:  
    print(i)
```

d). All of these

Select one alternative:

☐ d



☐ a

☐ c

☐ b

Maximum marks: 1

We have a dictionary as follows:

```
dictionary = {'student':["aparna", "pankaj", "sudhir", "Geeku"],
              'degree': ["MBA", "BCA", "M.Tech", "MBA"],
              'grade': ['B', 'B', 'A', 'C']}
```

Which of the following implementation would give the output as:

student	degree	grade
aparna	MBA	B
sudhir	M.Tech	A

IMPLEMENTATIONS ->

a).

```
df = pd.DataFrame(dictionary, index = [True, False, True, False])
print(df.loc[True])
```

b).

```
df = pd.DataFrame(dictionary, index = [0, 1, 2, 3])
print(df[[True, False, True, False]])
```

c).

```
df2 = pd.DataFrame(dictionary, index = [0, 1, 2, 3])
institute = np.array(['UiS', 'UiA', 'UiS', 'UiB'])
print(df2[institute == 'UiS'])
```

d). All of these

Select one alternative:

☐ d



☐ c

☐ a

☐ b

Maximum marks: 1

We have an array, `arr = np.arange(24).reshape(2,3,4)`. If we want to transpose on axis 0 and 1 keeping axis 2 unchanged,

we can use

`arr.transpose((1,0,2))`

Which of the following gives the same result?

Select one alternative:

☐ `arr.transform(0,1,2)`

☐ `arr.reshape(0,1,2)`

☐ `arr.swapaxes(0,1)`

☐ `arr.T(0,1)`



Maximum marks: 1

Observe the following code and identify what will be the outcome?

```
import numpy as np
```

```
x = np.array([[0, 1],  
              [2, 3]])  
np.transpose(x)
```

OUTPUT ->

a). `array([[0, 2],
 [1, 3]])`

b). `array([[0, 1],
 [2, 3]])`

c). `array([[2, 3],
 [0, 1]])`

d). None of these

Select one alternative:

☐ c

☐ d

☐ b

☐ a



Maximum marks: 1

Amongst which of the following if the syntax is true?

IMPLEMENTATIONS ->

a). `if condition:`
 `#Will execute this block if the condition is true`

b). `if condition`
`{`
 `#Will execute this block if the condition is true`
`}`

c). `if(condition)`
 `#Will execute this block if the condition is true`

d). None of these.

Select one alternative:

☐ d

☐ c

☐ a

☐ b



Maximum marks: 1

Observe the following code and identify what will be the outcome?

```
import numpy as np  
a = np.array([[1,2,3],[0,1,4]])  
  
print (a.size)
```

Select one alternative:

- ☐ 1
- ☐ 4
- ☐ 5
- ☐ 6



Maximum marks: 1

Observe the following code and identify what will be the outcome?

```
import numpy as np
a = np.array([[1,2,3],[0,1,4]])
b = np.zeros((2,3), dtype=np.int16)
c = np.ones((2,3), dtype=np.int16)
d = a + b + c

print (d[1,2])
```

Select one alternative:

☐ 5



☐ 4

☐ 7

☐ 3

Maximum marks: 1

Which of the following would extract all the first 3 rows of the last 5 columns in a given numpy 2d array 'a'?

Select one alternative:

☐ a[:3, -5:]



☐ a[:3, :5]

☐ a[:3, 5:]

☐ a[:3, :-5]

Maximum marks: 1